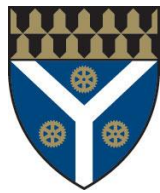




Automated Essay Scoring of College Application Essays with LLMs

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INTRODUCTION

Recent advances in Large Language Models (LLMs) have renewed interest in Automated Essay Scoring (AES) systems that can emulate or assist human evaluators. However, most existing AES systems are trained on academic writing or standardized tests, and fail to capture the narrative depth and emotional nuance central to college application essays.

This project investigates whether LLM-based systems can provide accurate and interpretable holistic scoring of Common App personal statements. Specifically, we ask: Can we model expert judgment of college essays using structured trait scores and transformer-based textual representations?

FINDINGS AND DELIVERABLES

> Codebase for AES pipeline

> Trait scores are positively correlated and align with overall ratings, but alone are poor predictors.

> Adding textual features (TF-IDF, BERT) improves performance ($R^2 = -0.0032$, $MSE = 1.0038$ to $R^2 = 0.1554$, $MSE = 0.8451$).

> Essays without low trait scores have higher median and ceiling ratings ($p < 0.01$).

BACKGROUND

Rather than relying solely on generic language features, we introduce and compute five targeted metrics that align with college consulting practices:

PST Ratio: How well an essay balances narrative structure across three key components: Problem, Solution, and Takeaway.

Specificity: The vividness and precision of an essay's language. High-scoring essays include concrete details, specific imagery, and nuanced descriptions.

Pace: How effectively the essay varies its sentence lengths to create dynamic storytelling.

VSPICE: How well the essay demonstrates six core character traits: Vulnerability, Selflessness, Perseverance, Initiative, Curiosity, and Expression.

Creativity: How original the essay's content and angle are compared to typical college application essays.

METHODOLOGY

This project uses a dataset of 1,117 Common App essays, written by real consenting students and provided by the college consulting firm ElevatEd. Using OpenAI's GPT-4o model, we assigned each essay scores (1.0–10.0) across five dimensions: PST Ratio, Specificity, Pace, VSPICE, and Creativity.

You are an expert college admissions counselor that has read over 10,000 essays. You receive a student essay and determine its Creativity score. The Creativity score is a number between 1.0 and 10.0 that measures how original the essay's content and angle are compared to typical college application essays. Creativity is based on the uniqueness of the story, the perspective, and the style of expression. Be thorough and harsh.

Score ranges:

- 1-3: Very common topic and angle; little originality in the content and angle
- 4-6: Somewhat common topic or approach, but some signs of unique thinking
- 7-8: Uncommon topic or a creative approach to a familiar idea
- 9-10: Highly original idea, story, or perspective rarely seen in other essays

The response must be the numerical score to one decimal place for the essay. The response must not contain any additional text.

Figure 1. Prompt for GPT-4o to calculate Creativity score

Our goal was to predict an essay's holistic rating by combining both prompt-engineered trait scores and essay embeddings. We extracted dense embeddings from each essay using a BERT encoder, concatenated them with the corresponding five trait scores, and fed the resulting feature set into regression models like Random Forest and Gradient Boosting.

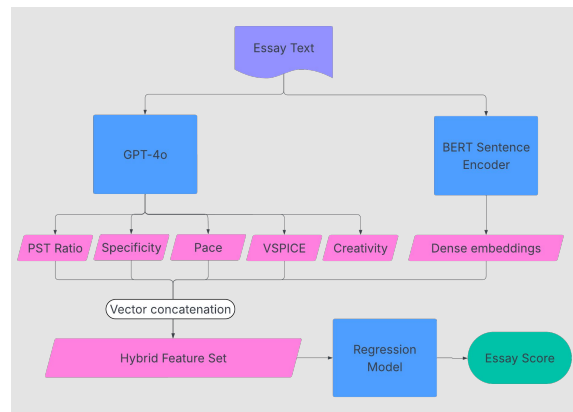


Figure 2. Model architecture using GPT-4o to extract metrics and BERT to generate dense embeddings.

RESULTS

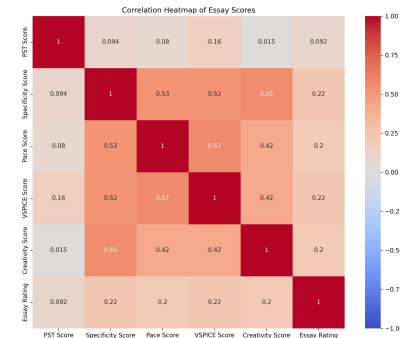


Figure 3. Correlation heatmap between trait scores and overall essay rating

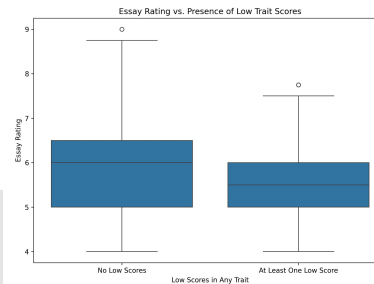


Figure 4. Boxplot comparing essay ratings for submissions with and without low trait scores

All five engineered traits are positively correlated with each other and with overall essay ratings, suggesting that strength in one area often reflects broader essay quality. However, the correlation with human-assigned ratings is relatively weak (≈ 0.2), indicating that these metrics only partially capture what reviewers value.

Essays without weak traits showed a higher median and a higher ceiling in overall ratings. We performed a two-sample t-test, which confirmed that the difference in overall ratings between the two groups was statistically significant ($p < 0.01$), reinforcing the value of trait scores as indicators of essay quality.

CONCLUSION

Trait scores show moderate correlation with overall ratings but alone lack predictive power. Yet adding textual features improves performance, underscoring the value of semantic signals. Essays with even one low trait score tend to receive significantly lower overall ratings, suggesting that these metrics can serve as effective flags for identifying weaker or unbalanced essays.

Future work could focus on refining GPT-4o prompts for more reliable trait scoring and exploring additional metrics, such as the strength of an essay's hook and conclusion.